SpikeShot:
Bio-Inspired Coincidence Gating for Neuromorphic Gunshot Detection

Muhammad Aitsam, ***Syed Saad Hasan***, *Alessandro Di Nuovo*

ICANN 2026 (International Conference on Artificial Neural Networks)

Centre of Excellence in AI and Robotics,
Sheffield Hallam University
Department of Computer, Control, and Management Engineering (DIAG),
Sapienza Università di Roma

Sheffield Hallam University | Centre of Excellence in Artificial Intelligence and Robotics

SAPIENZA UNIVERSITÀ DI ROMA

PRIMI
Performance in Robots Interaction via Mental Imagery

SIT

DIAG

The Problem: Deployed Gunshot Detection

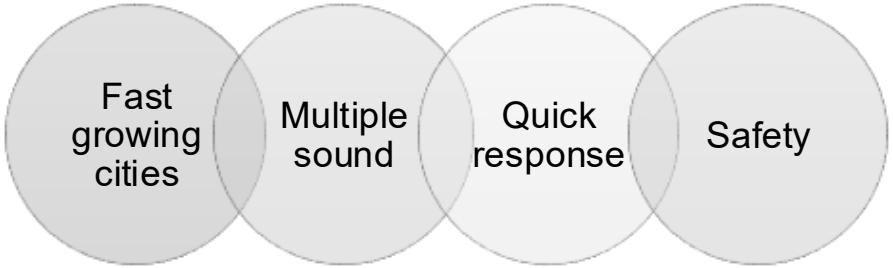
- 43,000+ firearm deaths annually in the United States.
- 3,511 incidents reported in South East Europe in 2025; domestic violence most frequent context.
- 68% increase in gun crime between 2024 and 2025.
- Economic impact: $557 billion annually; $2.1 million per homicide.
- Acoustic sensor networks deployed globally; commercial platforms (**ShotSpotter**) cost $65,000 per square mile per year.
- **Current approach:** Convolutional neural networks achieved approximately 90% accuracy with 15% false positive [3]. Laboratory conditions: 97–99% accuracy with ResNet18 and ensemble architectures [1,2].
- Gun violence remains a critical global public health crisis

Source:
https://www.commonwealthfund.org/publications/2024/oct/comparing-deaths-gun-violence-us-other-countries
https://www.seesac.org/News-SALW/snapshot-of-firearm-incidents-in-south-east-europe--2025/
https://globalinitiative.net/analysis/berlin-europe-trend-rising-criminal-violence/
https://manhattan.institute/article/thinking-through-the-shotspotter-debate

[1] Khan (2025). [2] Morehead et al. (2019). [3] Katsis et al. (2022).

The Solution: Spiking Neural Networks (SNNs)

THE URBAN CHALLENGE:



FOR A SAFER ENVIRONMENT:

Solution that bring all these variable together

INTELLIGENT | AUTOMATED | QUICKLY | ACCURATELY

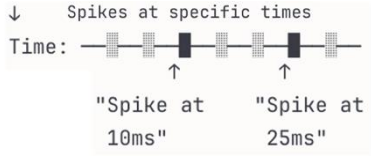
Why SNN?

Event Driven, Spike timing, Responde to changesSparse Processing, Can Detect with First few Spikes

EVENT-DRIVEN (Perfect for Gunshots/ Impulsive Audio):

```
Time: [0.0s]—[1.0s]—[2.0s]—[2.5s]—[3.0s]—[4.0s]
        ↓       ↓       ↓       ↓       ↓       ↓
      sleep   sleep   sleep   SPIKE!  sleep   sleep
```

SNN NATURALLY KNOW WHEN THING HAPPEN:



ENERGY EFFICIENCY:

```
Only processes spikes (maybe 1000 total in 1 hour)
Energy: ███ LOW
```

Datasets: Multi-Firearm and UrbanSound8K

Positive class: Gunshots:

Multi-firearm dataset (2,148 files) [10] + UrbanSound8K (374 gunshots) [11].

Negative class: Urban Sounds (Hard Negatives):

Eight UrbanSound8K categories (car horn, dog bark, jackhammer, drilling, siren, children playing, engine idling, street music) providing 2,522 balanced samples.

Items	Total Samples	Gunshots	Non-Gunshots	Balance
Over Samples	5,044	2,522	2,522	50% - 50%
Preprocessing	9,354	4,677	4,677	50% - 50%

Table 1

Preprocessing and splits:

Sample rate: 22.05 kHz, mono, peak-normalized. Training clips: 0.75 s (0.20 s before event, 0.55 s after).
File-level stratified splits: training 3,530, validation 757, test 757.
No file-level leakage. Validation/test: full 4.0 s for sliding-window evaluation.

[10] Kabealo et al. (2023). [11] Salamon et al. (2014).

Octopus Cells: Natural Model for Transient Detection

Octopus cells in the ventral cochlear nucleus are coincidence detectors tuned for broadband transients. These properties align with gunshot acoustics:

- Coincidence detection: synchronized input across tonotopic channels within ~1 millisecond [5]
- Depolarization rate sensitivity: dP/dt > 5–15 mV/ms, suppressing slow backgrounds [6]
- Low input resistance (7–10 MΩ) prevents single-channel triggering [7]
- Sub-200 microsecond spike-timing precision [8]
- Dendritic compensation for cochlear travelling-wave delays [9]

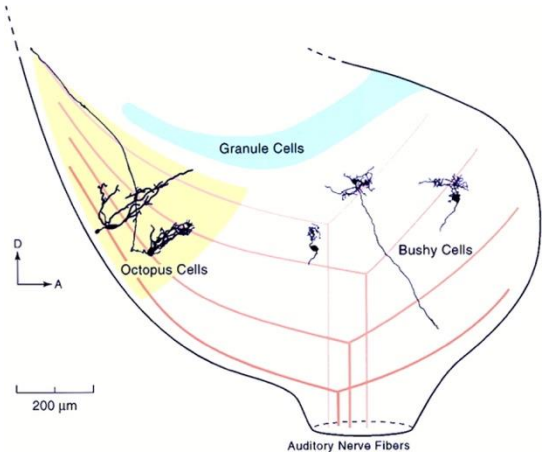


Figure from the Oertel, D. (1997). *Neuron*, 19(5), 959–962.

[5] Oertel et al. (2000). [6] Ferragamo & Oertel (2002). [7] Golding et al. (1995). [8] Golding et al. (1999). [9] McGinley et al. (2012*).*

Coincidence-Gated Temporal Contrast (CGTC)

$$\Delta M[f,t] = \log(M[f,t]) - \log(M[f,t-1])$$

$$\theta = \mu(\Delta M) + \sigma(\Delta M) \text{ Gate: } s[t]=1 \text{ if } (1/F)\Sigma f\ 1[\Delta M[f,t]>\theta] \geq \tau c=0.48$$

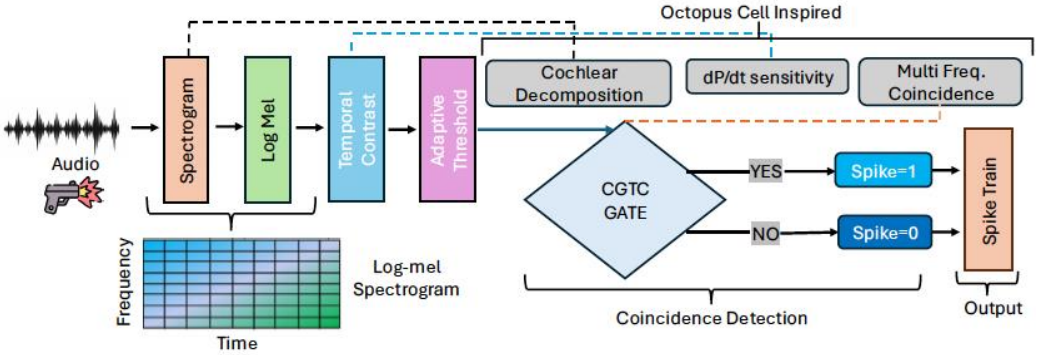


Figure 1 CGTC encoding pipeline

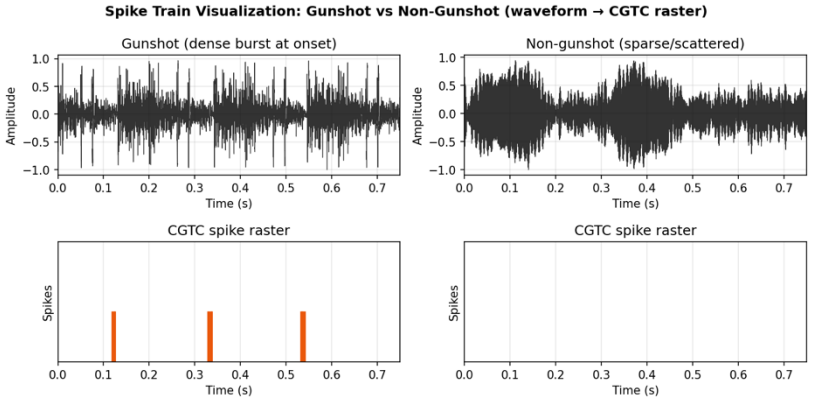


Figure 2

KEY PROPERTIES:

- 70% of non-gunshots produce zero spikes.
- Sparse activity in broadband transients: jackhammer (47.1%), car horn (24.9%), siren (19.6%)
- 128-fold dimensionality reduction (128 bins → 1D binary sequence)

Three SNN Architectures

We evaluate three architectures, all using leaky integrate-and-fire (LIF) neurons with membrane decay β = 0.85

- **HierarchicalSNN (29K parameters):**

  Balances sensitivity and specificity.

- **MILSNN (29K parameters):**

  Best overall accuracy and detection sensitivity

- **TemporalConvSNN (48K parameters):**

  Achieves lowest false positive rate (1.27%) and highest precision

All models trained with the same learning r*ate, weight decay, 50 epochs, focal loss (α = 0.25, γ = 2.0), and dropout (0.3).*
We report results over 5 random seeds (42–46).

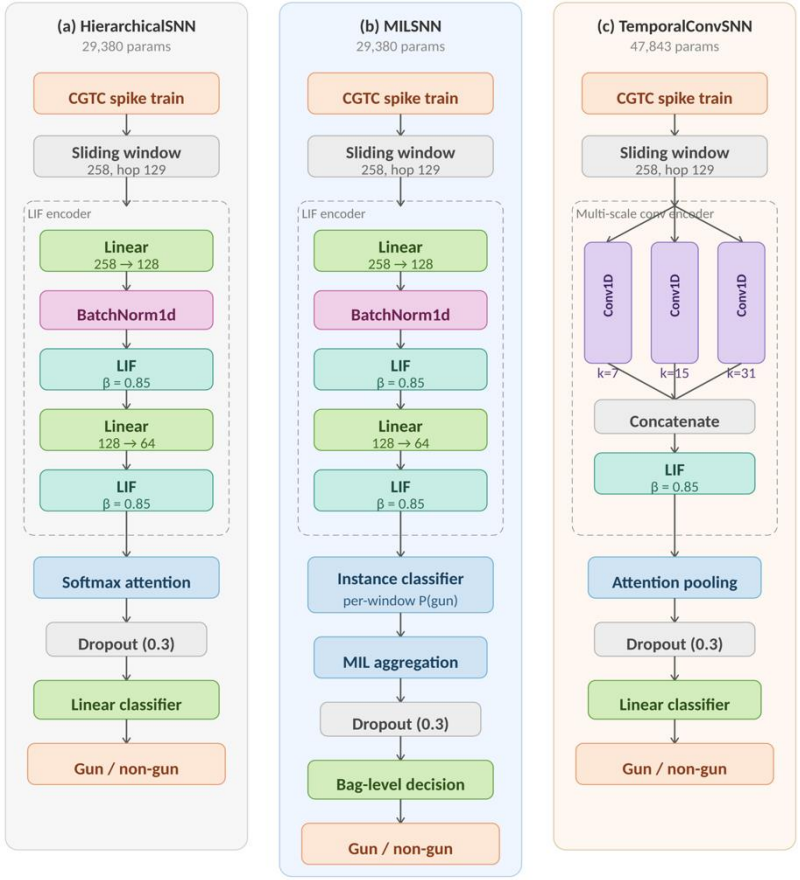
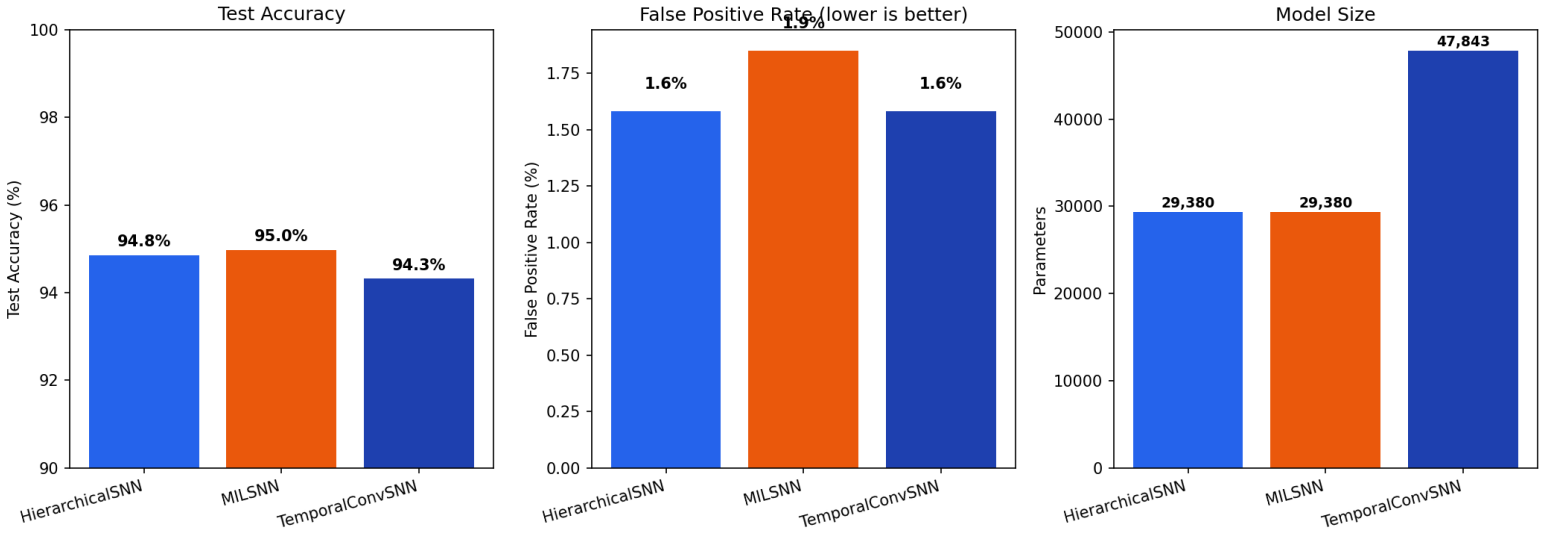


Figure 3

Results: Event-Centric Encoding (n=757)

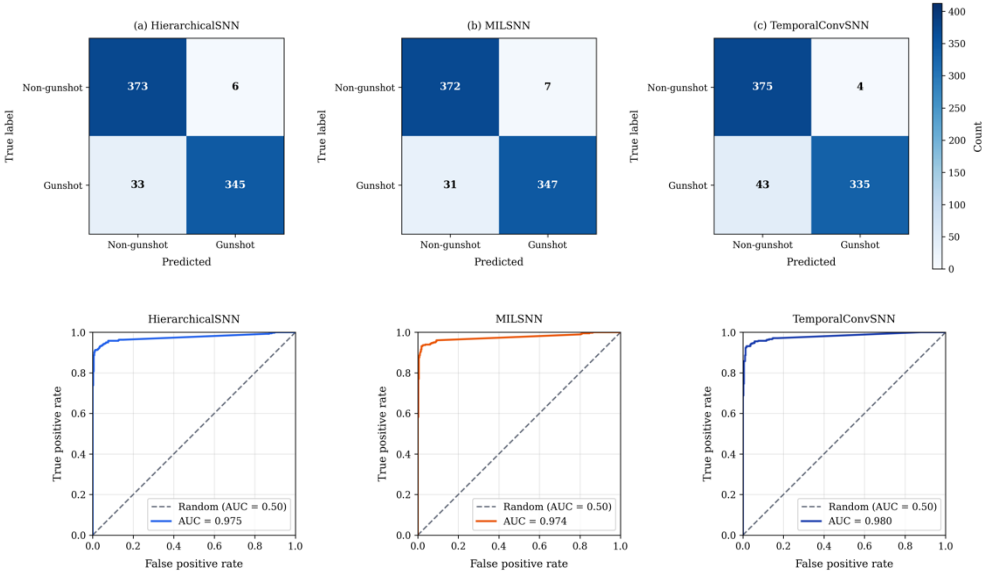


All three architectures exceed 93% accuracy, confirming that CGTC encoding preserves sufficient discriminative information for reliable classification.

Results & Critical Finding: Performance and Temporal Sensitivity

- MILSNN provides the best overall detection balance, only 7 false positives
- TemporalConvSNN produces fewest false alarms (4) but misses gunshots (43 false negatives).

- Temporal position of the encoding window has a substantial effect on classification performance.
- Sliding-window over full 4-second recording and aggregating maximum probability drops accuracy to 83.09%, an 11.52 percentage point gap.
- SNN pipeline must provide approximate event timing.



All models show strong discrimination with **AUC ≈ 0.97–0.98**.

Comparison with Published Methods

- Direct comparison across studies is limited because of differences in protocols
- CNN baselines: 97% (lab) [12], ~90% with 15% FPR (field) [13,14].
- TemporalConvSNN: 1.27% FPR with 400× fewer parameters.

Method	Type	Environment	Accuracy	FPR	Params	Notes
CNN-based methods						
ResNet18 [12]	CNN	Laboratory	99.0%	N/A	11.7M	Controlled dataset
CNN Ensemble [13]	CNN	Laboratory	99.0%	N/A	~1M	Multi-model inference
CNN (Field) [14]	CNN	Real-world	~90%	15.0%	~5M	Tropical forest recordings
SNN-based methods						
MILSNN (ours)	SNN	Balanced	**94.61%**	**2.85%**	**29K**	400× smaller; first SNN work
TemporalConvSNN (ours)	SNN	Balanced	**93.79%**	**1.27%**	**48K**	12× lower FPR vs field CNN

Table 2

[12] Khan (2025). [13] Morehead et al. (2019). [14] Katsis et al. (2022).

Contributions:

- First SNN application to gunshot detection with bio-inspired CGTC encoding.
- 94.61% accuracy, 1.27% FPR (12× lower than field CNN) with 29–48K parameters (400× smaller than ResNet18).
- Temporal alignment critical for SNNs: 11.52 pp accuracy gap reveals deployment constraint absent in CNNs

Future Work:

- 4.5–5.6× lower energy on neuromorphic hardware → battery-powered monitoring.
- Field-realistic FPR reduces alert fatigue; foundation for impulsive audio detection beyond gunshots.

Sheffield Hallam University | Centre of Excellence in Artificial Intelligence and Robotics

Sapienza Università di Roma — DIAG

PRIMI — Performance in Robots Interaction via Mental Imagery

Questions & Discussion
Thank you for your attention!

SpikeShot: Bio-Inspired Coincidence Gating for Neuromorphic Gunshot Detection

Muhammad Aitsam, **Syed Saad Hasan,** Alessandro Di Nuovo





Thank you ICANN 2026 and team